

Repository Appendix E: Extended Interpretation and Limitations for “Benchmark-Gated Composite Mean Estimation: Regime-Conditional Improvement and Its No-Free-Lunch Limits”

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1 Relationship to the journal submission

2 This repository appendix accompanies the article and preserves the ex-
3 tended interpretation, deployment discussion, limitations, and future re-
4 search agenda. The journal supplementary file retains Appendices A–C,
5 while detailed result narratives are provided in Repository Appendix D.
6 The version-controlled project is available at [https://github.com/ALRIER/](https://github.com/ALRIER/Genetic_Algo_Hull_University)
7 **Genetic_Algo_Hull_University**. This document is interpretive support
8 and does not replace the evidence or methods reported in the article and
9 journal supplementary material.

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Repository Appendix E: Extended interpretation, deployment, and limitations

E.1 Extended discussion

The main theoretical question in this study is not whether evolutionary search can generate a composite estimator, but under which conditions a weighted estimator can improve on strong classical and modern robust benchmarks. The results suggest that such improvement becomes possible when the distribution creates enough variance, tail instability, or contamination sensitivity for the sample mean and individual robust procedures to leave an exploitable region between efficiency and robustness. The genetic algorithm can search for that region, but it cannot create it. A composite succeeds only when the reduction in variance and extreme loss is large enough to compensate for the finite-sample bias introduced by components that do not individually target the arithmetic mean.

This interpretation follows directly from the MSE dominance condition developed in Section 2 of the main text. For a fixed weight vector w , improvement requires

$$B_n(w, F)^2 + V_n(w, F) < \text{Var}_F(\bar{X}_n).$$

The inequality explains both the successful and unsuccessful cases. Under right skewness, heavy tails, and contamination, the variance and upper-tail loss of the sample mean may become sufficiently large for a weighted combination to accept a limited increase in bias and still reduce total MSE. This is consistent with the classical robustness literature, which shows that small amounts of contamination can greatly affect moment-based estimators, and with modern work showing that robust mean estimation depends on assumptions about tails, variance, and contamination structure (Huber, 1964; Huber and Ronchetti, 2009; Hampel et al., 1986; Wilcox, 2012; Maronna et al., 2019; Catoni, 2012; Devroye et al., 2016; Lugosi and Mendelson, 2019).

The confirmed Lognormal results fit this mechanism particularly well. Lognormal distributions combine positive support, substantial right skewness, and multiplicative dispersion (Aitchison and Brown, 1957). These properties can produce a long upper tail in the sampling distribution of the mean, especially after upper-tail contamination. In such settings, robust and partially robust components can reduce the contribution of extreme observations, while a remaining mean component helps keep the composite aligned with μ_F . The confirmed Lognormal specialists are thus consistent with a

regime in which the variance reduction obtained from trimming, bounded influence estimation, or median-based components exceeds the squared bias introduced by those same components.

Statistical confirmation and practical significance are separate decisions. The benchmark gate used in this study asks whether the frozen composite improves on both criteria, so its evidential threshold is zero gain. No universal minimum deployment gain was prespecified. A practical deployment policy should add an application-specific utility gate, consistent with selective prediction and deferral under uncertainty (Geifman and El-Yaniv, 2017; Mozannar and Sontag, 2020):

$$\underline{G}_{\text{mean},k} > \tau_{\text{mean}}, \quad \underline{G}_{q_{0.95},k} > \tau_{q_{0.95}}, \quad \hat{U}_k > 0,$$

where the lower confidence bounds $\underline{G}$ must exceed thresholds fixed before deployment and $\hat{U}_k$ incorporates implementation, monitoring, and misclassification costs. Under this distinction, CV-010 is a statistically confirmed but operationally modest specialist. Its 2.2% $q_{0.95}$ gain establishes reproducible improvement under the study design, but it does not by itself justify the added complexity of profile matching and composite computation in every application.

This result also clarifies why target compatibility matters. The median, mode, geometric mean, and harmonic mean do not generally estimate the arithmetic mean under asymmetric distributions. Their contribution can still be useful in a finite-sample composite, but only when their stabilizing effect is large enough to improve MSE for the fixed target. Gneiting’s account of elicitable point functionals emphasizes that evaluation must remain tied to the stated target (Gneiting, 2011). In the present study, a component was not rewarded merely for being stable or robust. It contributed value only when the resulting composite estimated μ_F more accurately than the strongest admissible benchmark.

The same argument explains why large discovery gains did not always survive confirmation. Several Inverse Gaussian, Weibull, and Ex Wald candidates improved on the original classical benchmark set, but the margin disappeared after Catoni-type, median-of-means, winsorized, and tuned Huber estimators entered the comparison. These procedures already address many of the weaknesses of the sample mean under heavy tails and contamination (Catoni, 2012; Devroye et al., 2016; Lugosi and Mendelson, 2019, 2021; Minsker, 2023; Wilcox, 2012). Once they became available as competitors, the residual room for improvement became much smaller.

81 This finding supports an important condition for composite success: an
 82 estimator produced by the genetic algorithm can outperform a benchmark
 83 only when its combined influence profile captures useful structure that the
 84 benchmark does not already exploit. A mixture has no automatic advantage
 85 over a strong single estimator. When Catoni-type, median-of-means, tuned
 86 Huber, or winsorized estimators already provide an effective balance between
 87 bias and tail control, an additional convex combination may offer little bene-
 88 fit. The reduced accepted set under modern validation is thus evidence that
 89 the benchmark gate imposed a meaningful statistical standard.

90 The negative control results reinforce this interpretation. Under clean
 91 or near-efficient Normal settings, the sample mean has little excess variance
 92 available for a composite to remove. In that region, adding robust compo-
 93 nents can introduce bias or efficiency loss without generating enough protec-
 94 tion to compensate. This matches the expectation from classical estimation
 95 theory that the sample mean is difficult to improve under Gaussian squared
 96 error conditions (Lehmann and Casella, 1998; van der Vaart, 1998). The
 97 continued retention of the benchmark in Normal regimes is not an incidental
 98 outcome. It is evidence that the dominance region of a nontrivial composite
 99 contracts when F approaches a distribution under which the mean is already
 100 efficient.

101 The Ex Gaussian results show that skewness alone is not sufficient. Ex
 102 Gaussian distributions can be asymmetric and long tailed, yet no transfer-
 103 able composite emerged in the simulation controls. The benefit of weighting
 104 depends on more than a broad family label. It depends on the complete
 105 regime: contamination mechanism, contamination rate, outlier scale, sample
 106 size, and the relationship between these factors and the influence profiles of
 107 the available components. A family can be difficult in general and still offer
 108 no stable composite advantage once a strong benchmark is available. This
 109 finding argues against treating skewness or heavy tails as sufficient activation
 110 rules.

111 The expanded-basis experiment directly tested another theoretical ex-
 112 pectation: the set of attainable composites depends on the estimator library.
 113 Research on aggregation, model averaging, stacking, and the Super Learner
 114 shows that a learned combination is constrained by the candidate procedures
 115 available to it (Yang, 2001; van der Laan et al., 2007; Claeskens and Hjort,
 116 2008; Hoeting et al., 1999). The same principle appears in evolutionary com-
 117 putation, where representation defines the solutions that a search procedure
 118 can express (Bäck, 1996; Eiben and Smith, 2015; Telikani et al., 2021; Katoch

119 et al., 2021). A genetic algorithm cannot assign weight to an estimator that
 120 is absent from its chromosome.

121 The shift from Lognormal specialists under the ten-component basis to
 122 Weibull specialists under the 26-component basis is consistent with this prin-
 123 ciple. Adding Catoni-type, median-of-means, winsorized, and tuned Huber
 124 components did not simply increase the number of available coordinates. It
 125 expanded the set of possible influence profiles, bias patterns, and covariance
 126 structures that could be formed inside the simplex. The new Weibull spe-
 127 cialists could combine modern robust components with classical summaries
 128 in ways that were mathematically unavailable to the initial search.

129 This does not imply that a larger basis must always perform better. A
 130 richer library also creates a larger search space, increases the number of
 131 weak or redundant combinations, and can make useful regions more difficult
 132 to locate. Aggregation theory does not guarantee that every expansion of the
 133 library improves realized performance. It states that the achievable solution
 134 depends on what the library contains and on how effectively the learning
 135 rule navigates it (Yang, 2001; van der Laan et al., 2007; Claeskens and Hjort,
 136 2008). The expanded stage supports the first part of this claim because the
 137 specialist map changed. The Dirichlet audit also speaks to the second part
 138 because it identified useful regions that the completed genetic algorithm run
 139 had not reached.

140 The Weibull specialists reveal another central condition: estimator per-
 141 formance depends on the full data-generating regime, not only on distri-
 142 butional family. Their weights produced stable gains under locked-unseen
 143 regimes with extreme contamination but did not retain an advantage under
 144 the original-regime evaluation. In the language of the MSE decomposition,
 145 the same vector w encountered different values of $B_n(w, F)$ and $V_n(w, F)$ af-
 146 ter the regime changed. A weighting structure that reduced extreme variance
 147 enough to dominate under one contamination profile could retain too much
 148 bias, or achieve too little variance reduction, under another.

149 This result is not evidence of a general Weibull estimator. It identifies a
 150 narrower dominance region defined by Weibull structure together with severe
 151 upper-tail or point-mass contamination. The family label alone is insufficient
 152 for deployment. Activation must depend on a more detailed profile that in-
 153 cludes the contamination pattern and scale. This is one of the most important
 154 findings of the study because it moves the idea of regime dependence from a
 155 broad conceptual claim to an operational restriction.

156 The asymmetry between matched benefit and mismatched loss requires

157 a conservative pre-activation rule. The benchmark remains the default. A
158 specialist may be considered only when an independent diagnostic, specified
159 without using the candidate’s observed performance, supports the validated
160 family, contamination direction and structure, severity range, and sample-
161 size range. Activation must also clear the practical utility gate above. If the
162 profile is ambiguous, falls near a boundary, changes during use, or lacks a
163 validated diagnostic mapping, the procedure abstains and retains the bench-
164 mark, following the established logic of reject-option and learning-to-defer
165 systems (Geifman and El-Yaniv, 2017; Mozannar and Sontag, 2020). One
166 possible utility representation is

$$\hat{U}_k = (1 - \hat{p}_{\text{mis}})\hat{G}_k - \hat{p}_{\text{mis}}\hat{L}_k - C_k,$$

167 where $\hat{p}_{\text{mis}}$ is the estimated probability of profile misclassification, $\hat{G}_k$ is the
168 expected matched-regime benefit, $\hat{L}_k$ is the expected mismatched-regime loss,
169 and C_k is the implementation cost. The current study does not estimate $\hat{p}_{\text{mis}}$
170 or validate an autonomous real-time classifier. The specialists are there-
171 fore validated statistical configurations, not autonomous deployment rules.
172 The large negative original-regime outcomes and the full candidate matrix
173 in Supplementary Table S3 in the journal supplementary material make this
174 restriction explicit.

175 The difference between discovery in one regime and confirmed gain in a
176 related locked-unseen regime is also informative. Evolutionary discovery can
177 identify a region of the simplex whose usefulness extends beyond the exact
178 cells that produced it, but the location of its strongest advantage may shift
179 within the surrounding structural neighborhood. In this sense, discovery
180 does not necessarily recover the single best vector for one fixed cell. It can
181 uncover a weighting pattern associated with a broader local geometry of
182 related regimes. Confirmation then identifies where within that geometry
183 the frozen vector actually satisfies the dominance condition.

184 The robustness audit helps explain how the confirmed Weibull composites
185 achieved this behavior. Both specialists retained positive weight on the sam-
186 ple mean, so their formal breakdown point remains zero. Classical influence-
187 function theory makes clear that an unbounded component with positive
188 weight transfers part of its unbounded influence to the composite (Hampel
189 et al., 1986; Huber and Ronchetti, 2009; Avella-Medina, 2017). Yet the mag-
190 nitude of that contribution was strongly reduced. The estimated effect of one
191 arbitrarily large observation was approximately 1/62 and 1/46 of its effect
192 on the sample mean.

193 This distinction matters. A breakdown point of zero does not mean
 194 that the composite behaves identically to the sample mean under practi-
 195 cally relevant contamination. Formal breakdown describes what can happen
 196 under arbitrarily severe replacement contamination. Gross error sensitivity
 197 describes how strongly a small contamination perturbation can affect the esti-
 198 mator (Hampel et al., 1986; Huber and Ronchetti, 2009). The two specialists
 199 did not acquire full high-breakdown robustness, but they substantially atten-
 200 uated the influence of extreme observations while preserving some alignment
 201 with the arithmetic mean. Their gains are consistent with this intermediate
 202 position between efficiency and robustness.

203 The composition of the weights supports the same interpretation. Most
 204 of the mass was assigned to bounded or controlled influence procedures, in-
 205 cluding Huber, Catoni-type, biweight, median, and winsorized estimators.
 206 Modern work on M estimators and robust means shows how bounded influ-
 207 ence and deviation control can stabilize estimation under contamination and
 208 heavy tails (Mathieu, 2022; Catoni, 2012; Wang and Ramdas, 2023; Lugosi
 209 and Mendelson, 2019). The small positive mean weight prevented complete
 210 separation from the target, while the robust components reduced exposure
 211 to the upper tail. The resulting estimator was not maximally robust in a
 212 formal sense; it was tuned to reduce risk for a specific regime.

213 The confirmation audits also address whether the observed dominance
 214 could be explained by random seed selection or post-selection optimism. The
 215 agreement between pooled and split-sample estimates indicates that the gains
 216 did not depend materially on using the same complete validation collection
 217 for selection and reporting. The survival of the specialists across eight val-
 218 idation seeds and under false discovery rate correction further supports the
 219 stability of the signal (Benjamini and Hochberg, 1995). These checks do not
 220 create the advantage, but they reduce the plausibility that it arose from an
 221 isolated Monte Carlo realization.

222 The bootstrap audit introduces a narrower caveat. Ordinary bootstrap
 223 intervals can behave poorly for medians and mode-based procedures (Ghosh
 224 et al., 1984; Romano, 1988). Modal weight was negligible in the confirmed
 225 Weibull specialists, which limits its practical impact on the reported inter-
 226 vals. Median weight was more visible in one specialist, so the finite-sample
 227 approximation may still converge more slowly than it would for a fully smooth
 228 estimator. This issue does not overturn the current result, but it identifies
 229 a specific extension: future confirmation can compare the ordinary paired
 230 bootstrap with m out of n subsampling when nonsmooth components re-

231 ceive substantial weight.

232 The random Dirichlet audit also changes how abstention should be inter-
233 preted. In many cells, 4,000 independent simplex draws failed to produce a
234 vector that passed both MSE gates across all seeds. Those results indicate
235 that no stable gate-passing configuration was found under the tested ba-
236 sis, Dirichlet proposal distribution, and audit budget. The recovery of both
237 Weibull specialists as positive controls, together with continued abstention in
238 all Normal and Ex Gaussian controls, shows that the audit could distinguish
239 known valid regions from benchmark-dominant settings.

240 At the same time, the audit found gate-passing configurations that re-
241 mained stable across all seeds in eight regimes not recovered by the com-
242 pleted genetic algorithm run. The strongest case, Q1R003, had a pass rate
243 of 1.8%, while the remaining rates were below 0.3%. These frequencies in-
244 dicate that successful configurations can be uncommon under the selected
245 Dirichlet proposal distribution. They do not by themselves establish the vol-
246 ume, connectivity, or geometry of the corresponding sets within the simplex.

247 The result also shows that benchmark retention can have two different
248 interpretations. In some regimes, the audit found no stable passing configu-
249 ration within its sampling budget. In others, passing configurations were
250 present but were not recovered by the completed genetic algorithm run.
251 Benchmark retention may thus reflect either limited evidence of an accessi-
252 ble gain under the audit distribution or incomplete exploration by the search
253 procedure.

254 This distinction limits what can be concluded about the genetic algorithm
255 itself. The validated specialists remain valid because their status depends on
256 frozen evaluation, not on the claim that the GA is the best optimizer. The
257 Dirichlet results suggest that alternative mutation schedules, larger popula-
258 tions, longer persistence, CMA-ES, mirror descent, or other simplex methods
259 could improve discovery efficiency (Beck and Teboulle, 2003; Hansen, 2016;
260 Bergstra and Bengio, 2012). Such comparisons would address how effectively
261 the search finds candidate regions. They would not change the benchmark
262 gate that determines whether a persisted vector deserves confirmation.

263 The external evaluation provides a final test of the regime matching hy-
264 pothesis. The frozen specialists achieved corrected wins across multiple inde-
265 pendent parent datasets and repeated subregimes without retraining. These
266 results indicate that some weighting patterns learned in synthetic regimes can
267 remain useful when applied to empirical distributions with compatible pro-
268 files. The use of independent parent datasets as the main unit of evidence

269 also avoids counting nested evaluations as independent replications (Hurl-
270 bert, 1984; Aarts et al., 2014).

271 The external signal was selective. It appeared in Lognormal, Normal, and
272 Weibull profiles and was absent from Ex Gaussian, Ex Wald, and Inverse
273 Gaussian profile-eligible evaluations. This pattern confirms that simulation
274 success does not map mechanically onto every empirical family. It also shows
275 that a specialist discovered under one synthetic condition may contribute
276 to an empirical profile only when the observed distribution, contamination
277 pattern, and benchmark relationship are sufficiently compatible.

278 The appearance of external wins in Normal profiles deserves careful inter-
279 pretation because Normal regimes served as negative controls during simula-
280 tion. The result does not necessarily indicate a contradiction. The external
281 profile classification captures empirical similarity, not exact Gaussian sam-
282 pling under the clean simulation condition. A dataset assigned to a Normal
283 profile may still contain finite-sample irregularities, endogenous contamina-
284 tion, heterogeneity, or tail behavior that creates room for a specialist to
285 improve recovery of the full-sample reference mean. The correct conclusion
286 is that the activation rule must be based on the complete empirical profile,
287 not the family label in isolation.

288 The external target also limits the claim. The full-sample mean served
289 as the reference because the true population mean is unknown in public
290 observational data. The external analysis thus evaluates recovery of an em-
291 pirical reference from repeated subsamples. It does not establish that the
292 same improvement would hold against an independently known population
293 parameter. Even within this restricted target, the consistency between sim-
294 ulation and external activation supports the underlying claim that estimator
295 performance depends on matching the weighting structure to the regime.

296 The findings align with the structural lesson of the No Free Lunch lit-
297 erature, although they do not constitute a proof of the formal optimiza-
298 tion theorem. Wolpert and Macready show that no optimizer dominates
299 when performance is averaged across unrestricted problem classes, while later
300 work emphasizes the role of inductive biases and restrictions on the environ-
301 ment (Wolpert and Macready, 1997; Wolpert, 2023; Sterkenburg and Grün-
302 wald, 2021; Goldblum et al., 2024). The present study expresses the same
303 logic in estimator selection: weights that are useful under one combination
304 of skewness, tail behavior, and contamination can lose their advantage when
305 those conditions change.

306 The results also show that conditional performance is not equivalent to

307 arbitrary post hoc specialization. The specialist configurations were sub-
 308 jected to frozen validation, new seeds, locked-unseen conditions, multiplicity
 309 control, negative controls, and external testing. These safeguards separate a
 310 genuine local dominance region from a candidate that only fits the discovery
 311 simulations. The benchmark gate is central because it permits abstention
 312 when the evidence does not support activation.

313 The broader methodological implication is that the GA should be under-
 314 stood as a generator of hypotheses over the estimator simplex. Statistical
 315 authority comes from the later comparison with admissible benchmarks. Un-
 316 der favorable regimes, evolutionary search can combine complementary influ-
 317 ence profiles and identify a vector that reduces both average and upper-tail
 318 loss. When the mean is near-efficient, the benchmark already captures the
 319 available gain, or the regime is mismatched, the same process should return
 320 no usable composite.

321 The evidence supports a clear set of conditions for success. A weighted
 322 estimator produced by the genetic algorithm is most likely to improve on the
 323 benchmark when the sample mean has substantial variance driven by the
 324 upper tail, no single robust benchmark fully captures the regime structure,
 325 the estimator basis can express a suitable balance between bias and influence,
 326 and the learned vector is applied within a compatible validation region. It is
 327 unlikely to improve when the mean is already efficient, when a modern robust
 328 benchmark absorbs most of the available gain, when the relevant component
 329 is missing from the basis, or when the deployment regime changes the balance
 330 between bias and variance.

331 In this sense, the main contribution is not a claim that evolutionary
 332 weighting produces a superior estimator in general. It is a tested account of
 333 when such weighting can be useful. The search identifies possible regions of
 334 advantage, the benchmark gate determines whether the gain is real, and the
 335 profile rule limits application to conditions supported by the evidence. Re-
 336 taining the benchmark outside those regions is part of the intended behavior
 337 of the method, not a failure of discovery.

338 *E.2 Extended limitations and future research*

339 The first limitation concerns the range of statistical environments rep-
 340 resented in the study. The simulation design assumes independent and
 341 identically distributed observations, is univariate, and focuses on sample
 342 sizes $n \geq 300$. This scope made it possible to isolate how distributional

343 shape, contamination mechanism, severity, and estimator composition af-
 344 fect the recovery of μ_F . It does not yet cover dependence between obser-
 345 vations, multivariate targets, relatively small samples, censoring, or changes
 346 in the data-generating process over time. These extensions are especially
 347 relevant for reaction time research, where trials are nested within partici-
 348 pants, individual distributions can differ substantially, and temporal effects
 349 such as learning, fatigue, and adaptation may violate simple iid assump-
 350 tions (Luce, 1986; Brenner and Smeets, 2019; Tejo et al., 2018; Chávez De la
 351 Peña and Vandekerckhove, 2025). Future work should examine hierarchical
 352 and participant-specific specialist maps, with validation performed at both
 353 trial and participant levels. Such designs would test whether a simplex re-
 354 gion learned from pooled observations remains useful after within-subject
 355 dependence and between-subject heterogeneity are represented explicitly.

356 The public data battery provides an important extension beyond simu-
 357 lation, but its target remains operational. Because the population mean is
 358 unknown in observational datasets, the full-sample mean was used as the
 359 reference for repeated subsample recovery. The external findings show that
 360 frozen specialists can recover this empirical target more accurately under
 361 matched profiles, but they do not establish performance against an indepen-
 362 dently known population parameter. Future external studies could use very
 363 large reference samples, administrative populations, controlled experimental
 364 archives, or prospective data collection to create a stronger approximation to
 365 the true target. In reaction time applications, this could include high-trial-
 366 count experiments in which a large participant-level distribution serves as
 367 the reference and smaller trial sets reproduce the practical conditions faced
 368 by applied researchers.

369 The empirical contamination rule also deserves further study. Public data
 370 irregularities were detected through the fixed $z = 3$ rule based on the median
 371 and unscaled MAD, with no synthetic contamination added. This choice
 372 supports a reproducible external evaluation, but the present study does not
 373 include a sensitivity analysis of the threshold. In right-skewed Lognormal
 374 or Weibull marginals, legitimate tail observations may be flagged as opera-
 375 tional irregularities even when they arise from the main distribution (Berger
 376 and Kiefer, 2021; Miller, 2023). The external results are consequently condi-
 377 tional on $z = 3$ and should not be interpreted as invariant to the diagnostic
 378 threshold. A focused extension should repeat the frozen external battery
 379 at $z \in \{2.5, 3, 4\}$ and compare eligible parents, profile-matched comparisons,
 380 CI-confirmed wins, survivors after false discovery rate correction, and family-

specific activation counts. Reaction time observations may contain fast guesses, attentional lapses, deadline effects, or participant-specific secondary processes, each of which can produce a different tail structure (Schwarz, 2001; Heathcote, 2004; Marmolejo-Ramos et al., 2023).

A second limitation concerns the discovery map. Candidate nomination rests on two discovery seeds, while confirmation uses eight additional seeds, higher Monte Carlo precision, paired bootstrap intervals, locked-unseen regimes, and multiplicity control. The confirmed claims do not depend solely on the two discovery streams, but the number and location of nominated candidates may still vary under a larger set of discovery seeds, a form of selection variability that can generate optimistic candidate estimates when discovery and assessment are not adequately separated (Cawley and Talbot, 2010; van Zwet and Cator, 2021). A natural extension is to estimate selection probabilities for each region of the simplex. Repeated discovery runs could show whether the same weight structures appear consistently, whether several equivalent solutions occupy a common neighborhood, or whether a candidate depends on an unusually favorable evolutionary trajectory. This would refine the map of discoverable regions without changing the frozen validation standard used to evaluate any persisted candidate.

The optimizer itself also remains open to comparison. The study uses a genetic algorithm because the objective is stochastic, nonsmooth, and constrained to the simplex, and because population-based search supports warm starts, mutation, and persistence across related regimes. The findings do not establish that the GA is the most efficient method for locating gate-passing regions. The Dirichlet audit showed that some useful areas may occupy very small portions of the simplex and may not be reached by a completed evolutionary run. Equal-budget comparisons with random search, CMA-ES, projected or mirror descent, Bayesian optimization, and alternative evolutionary designs would help determine which search strategy identifies these regions most efficiently (Bergstra and Bengio, 2012; Beck and Teboulle, 2003; Hansen, 2016; Bäck, 1996; Eiben and Smith, 2015). This comparison concerns discovery efficiency. The evidential status of a candidate would still depend on frozen weights, benchmark comparison, and post-discovery confirmation.

The expanded-basis results also motivate a more systematic study of estimator libraries. Adding Catoni-type, median-of-means, winsorized, and tuned Huber components changed which specialist regions could be expressed and produced the confirmed Weibull candidates. A larger library can create

new influence and bias combinations, but it can also introduce redundant coordinates and increase search difficulty (van der Laan et al., 2007; Claeskens and Hjort, 2008; Yang, 2001). Future studies should compare nested estimator bases and measure the marginal value of each component family. Such an analysis could determine whether a specialist depends on a particular modern robust estimator, on covariance between several components, or simply on the greater flexibility of the expanded simplex. Prespecified ablations of warm starts, inheritance, mutation concentration, and HPF exposure would complement this analysis.

Multiplicity control is concentrated at the stages where evidential claims are made. Discovery serves as a screening process and does not independently establish significance. Fixed-weight confirmation uses paired uncertainty estimates and a Benjamini-Hochberg correction across confirmation candidates, while the external battery applies false discovery rate control across profile-eligible comparisons (Benjamini and Hochberg, 1995). The correction reported in Section 4.4 of the main text left the confirmed set unchanged. Future studies could strengthen this design through a fully pre-registered confirmation list, independent simulation batches reserved before discovery, or sequential error control when candidates are evaluated over several waves (Robertson et al., 2023). These changes would provide an even sharper separation between exploratory search and confirmatory evidence.

The evidence taxonomy should also be developed prospectively. The internal categories summarize existing fixed-weight validation outcomes, but their detailed definitions were finalized after the broad simulation patterns were known. They are frozen and versioned for transparent reporting, yet they are not treated as retrospectively preregistered (Nosek et al., 2018). The external eligibility, benchmark policy, and false discovery procedure were specified before external wins were counted. Future work should apply the same prospective standard to the internal taxonomy, including explicit criteria for transfer specialists, local candidates, near-gate cases, and negative controls.

The Dirichlet audit introduces a further distinction that future studies should investigate directly. Benchmark retention can indicate that no stable gain exists in the evaluated regime, or that a useful region exists but occupies such a small part of the simplex that the completed search did not locate it. Persisting the strongest Dirichlet vectors and passing them through the full frozen validation procedure would test these two explanations. Q1R003 and the other flagged regimes offer concrete starting points.

457 This extension could also examine the local structure surrounding confirmed
 458 and audit-identified configurations. Systematic perturbations of the weights
 459 could determine whether successful vectors belong to broad neighborhoods
 460 of similar performance or to more localized sets requiring targeted search.
 461 The present Dirichlet audit identifies candidate configurations but does not
 462 estimate their geometry or volume within the simplex.

463 The Catoni-type components use a fixed implementation-specific sensi-
 464 tivity grid instead of the variance- and confidence-dependent calibration re-
 465 quired for Catoni’s original deviation guarantee. Their contribution should
 466 therefore be interpreted as empirically validated within the scales and regimes
 467 examined here, not as an implementation inheriting Catoni’s theoretical
 468 finite-sample bound. Scale-adaptive or plug-in calibration is an important
 469 extension for future work.

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